

Question 1:

① 100, ② 108, ③ 103, ④ 115, ⑤ 110

$$(a) R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$R_2 = \frac{108 - 100}{100} = 0.08$$

$$R_3 = \frac{103 - 108}{108} = -0.0463$$

$$R_4 = \frac{115 - 103}{103} = 0.1165$$

$$R_5 = \frac{110 - 115}{115} = -0.0435$$

$$(b) r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

$$r_2 = \ln\left(\frac{108}{100}\right) = 0.077$$

$$r_3 = \ln\left(\frac{103}{108}\right) = -0.0499$$

$$r_4 = \ln\left(\frac{115}{103}\right) = 0.1102$$

$$r_5 = \ln\left(\frac{110}{115}\right) = -0.0445$$

$$(c) (r_t \approx R_t)$$

$$\text{Month } 1 \rightarrow 2: |R_2 - r_2| = 0.003$$

$$\text{Month } 2 \rightarrow 3: |R_3 - r_3| = 0.0011$$

$$\boxed{\text{Month } 3 \rightarrow 4: |R_4 - r_4| = 0.0063} \rightarrow \text{least accurate month}$$

$$\text{Month } 4 \rightarrow 5: |R_5 - r_5| = 0.0010$$

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(1 + R_t)$$

Expansion



$$r_t = \boxed{R_t - \frac{R_t^2}{2}}$$

$$r_t \approx R_t$$

∴ If absolute diff. b/w r_t and P_t is larger → approximately least accurate

$$\left(\because R_t = \frac{P_t}{P_{t-1}} - 1 \right)$$

Question 2

$$E[R_1] = 15\% \quad , \quad E[R_2] = 7\%$$

$$\text{weight} : \underline{w_1 = 0.4} \text{ and } \underline{w_2 = 0.6}$$

(a)

$$E[ax + bY] = a E[X] + b E[Y] \quad \text{--- (1)}$$

$$\begin{aligned} E[R_p] &= E[w_1 R_1 + w_2 R_2] \\ &= w_1 E[R_1] + w_2 E[R_2] \\ &= 0.4 \times 0.15 + 0.6 \times 0.07 \\ &= 0.102 \end{aligned}$$

Expected portfolio return $\rightarrow$ 10.2%

(b)

No, the result won't change.

As, this relation in eqⁿ (1) holds true for all variables regardless if they are ~~interrelated~~ correlated or not.

Question 3

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

Let ~~Let~~ ^{Let} $X = w_1 R_1$, $Y = w_2 R_2$

$$R_p = w_1 R_1 + w_2 R_2$$

$$\therefore \text{Var}(w_1 R_1 + w_2 R_2) = \text{Var}(w_1 R_1) + \text{Var}(w_2 R_2) + 2 \text{Cov}(w_1 R_1, w_2 R_2)$$

Now, $\therefore$ $\boxed{\text{Var}(aX) = a^2 \text{Var}(X)}$, $\boxed{\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)}$

$$\text{Var}(w_1 R_1 + w_2 R_2) = w_1^2 \text{Var}(R_1) + w_2^2 \text{Var}(R_2) + 2 w_1 w_2 \text{Cov}(R_1, R_2)$$

$$= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_1 \sigma_2 \rho_{12}$$

$$\therefore \rho_{12} = \frac{\text{Cov}(R_1, R_2)}{\sigma_1 \sigma_2}$$

Question 4:

Asset A: $\mu = 13\%$, $\sigma = 25\%$

Asset B: $\mu = 8\%$, $\sigma = 12\%$

Portfolio weights: $w_A = 0.5$, $w_B = 0.5$

(a) σ_p for $\rho = +1, 0, -1$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho$$

$$= (0.5)^2 (0.25)^2 + (0.5)^2 (0.12)^2 + 2(0.5)(0.5)(0.25)(0.12)\rho$$

$$= 0.019225 + 0.015\rho$$

$$\left[\begin{array}{l} \rho = 1 \rightarrow \sigma_p^2 = 0.019225 + 0.015 = 0.034225 \\ \sigma_p = 0.185 = 18.5\% \\ \rho = 0 \rightarrow \sigma_p^2 = 0.019225 \\ \sigma_p = 0.13865 = 13.87\% \\ \rho = -1 \rightarrow \sigma_p^2 = 0.019225 - 0.015 = 0.004225 \\ \sigma_p = 0.065 = 6.5\% \end{array} \right.$$

(b) weighted avg. of individual risk: $w_A \sigma_A + w_B \sigma_B$

for $\rho = 1$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B (1)$$

$$\sigma_p^2 = (w_A \sigma_A + w_B \sigma_B)^2$$

$$\rightarrow \boxed{\sigma_p = w_A \sigma_A + w_B \sigma_B}$$

$\therefore$ for $\rho = 1$ $\Rightarrow$ σ_p

② $\rho = -1$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 - 2w_A w_B \sigma_A \sigma_B$$

$$\sigma_p^2 = (w_A \sigma_A - w_B \sigma_B)^2$$

To drive risk to zero: $\sigma_p = 0$

$$\Rightarrow (w_A \sigma_A - w_B \sigma_B)^2 = 0$$

$$w_A \sigma_A = w_B \sigma_B$$

$$w_A \sigma_A = (1 - w_A) \sigma_B$$

$$w_A (0.25) = (1 - w_A) (0.12)$$

$$\rightarrow w_A^* = \frac{0.12}{0.37} = 0.3243$$

Question 5:

Portfolio variance formula:

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho$$

Here all terms $w_A, \sigma_A, w_B, \sigma_B$ are > 0 .

When, $\rho < 1 \rightarrow$ overall value of σ_p^2 decreases.

($\because 2w_A w_B \sigma_A \sigma_B \rho < 2w_A w_B \sigma_A \sigma_B$)

than, when $\rho = 1$

Case of $\rho = 1$
 $\left[\begin{array}{l} \text{portfolio risk} = \text{wei} \\ \text{variance} = \text{portfolio} \\ \text{of } i \end{array} \right]$

Question 6:

Asset	μ_i	σ_i
1	15%	25%
2	8%	12%
3	5%	4%

$$\rho_{12} = 0.4$$

$$\rho_{23} = 0.2$$

$$\rho_{13} = 0.1$$

$$\rho_{ii} = 1$$

(a) Σ Diagonal: ($i=j$ and $\rho=1$)

~~Main diagonal~~

$$\Sigma_{11} = \sigma_1^2 = (0.25)^2 = 0.0625$$

$$\Sigma_{22} = \sigma_2^2 = 0.0144$$

$$\Sigma_{33} = \sigma_3^2 = 0.0016$$

off-diagonal:

$$\Sigma_{12} = \Sigma_{21} = \rho_{12} \sigma_1 \sigma_2 = 0.012$$

$$\Sigma_{13} = \Sigma_{31} = \rho_{13} \sigma_1 \sigma_3 = 0.001$$

$$\Sigma_{23} = \Sigma_{32} = \rho_{23} \sigma_2 \sigma_3 = 0.00096$$

$$\Sigma = \begin{bmatrix} 0.06250 & 0.01200 & 0.00100 \\ 0.01200 & 0.01440 & 0.00096 \\ 0.00100 & 0.00096 & 0.00160 \end{bmatrix}$$

Since all off diagonal elements are ^{symmetrically} equal $\rightarrow \Sigma$ is symmetric

Question 7 :

$$R_f = 4\%$$

Portfolio	$E(R_p)$	σ_p	Sharpe Ratio
X	14%	22%	?
Y	9%	10%	2
Z	20%	35%	?

(a)

$$\text{Sharpe Ratio (SR)} = \frac{E(R_p) - R_f}{\sigma_p}$$

$$(SR)_X = \frac{14\% - 4\%}{22\%} = 0.4545$$

$$(SR)_Y = \frac{9\% - 4\%}{10\%} = 0.5$$

$$(SR)_Z = \frac{20\% - 4\%}{35\%} = 0.4571$$

(b)

Ranking: $Y > Z > X$

Sharpe Ratio measures the ~~highest~~ excess return per unit risk taken, and investors would prefer highest SR

$\therefore Y$ would be preferred.

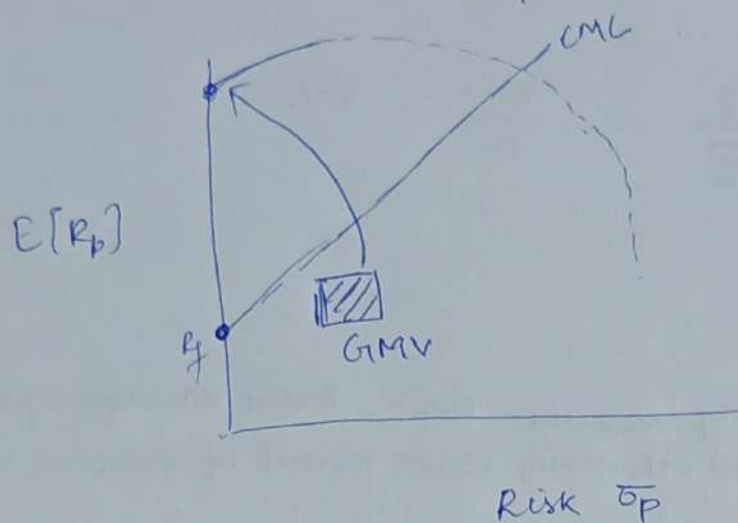
(c) An investor would prefer Y because it offers a strictly better risk-return tradeoff.

If an investor desires 20% expected return of portfolio Z, they don't need to accept its high risk of 35%.

Instead, they could choose Y and use R_f to scale their expected return to 20%, and that too in a much lower risk.

Question 3:

- (a) ~~Graph~~ GMV :- Combination of assets that result in lowest possible variance, and standard deviation, among all the possible portfolios in feasible set frontier



Here, moving further to left means decreasing risk, and since GMV mathematically represents min. possible value of σ , it must be a point close to y-axis, making it leftmost tip of frontier curve.

No, it's not correct, as the lower half may have very low risk, but in the upper half at the same level of risk, one can get higher expected return. Thus, upper half should be chosen.

CAL is the line from R_f , tangent to efficient frontier
Slope of CAL is Sharpe Ratio of market portfolio - the maximum risk-adjusted return achievable by an investor.

Question 9:

- (a) 36 observations, $\sigma_I^2 = 30\%$, $\hat{\mu}_i = ?$

$$\mu_i = \frac{\sigma_i}{\sqrt{36}} = \frac{30\%}{6} = 5\%$$

- (b) "Error maximisation" property describes how mean-variance optimiser blindly amplifies small estimation errors present in expected return.

~~Replacing~~ ML generates more accurate predictions for μ_i and ~~quantifying the~~

- (c) $N = 50$ assets

Σ contains $\frac{N(N+1)}{2}$ unique entries

$$\Rightarrow \frac{50 \times 51}{2} = \boxed{1275}$$

The challenge is that estimating this massive no. of parameters, ~~from historical samples~~ as the sample covariance matrix becomes unstable for large no. of assets.

Even a tiny fluctuation in data can cause optimiser to output wildly different and unreliable weights.